CLOSED IMMERSIONS APPEAR AT A FINITE STAGE UNDER AFFINE CLOSED TRANSITIONS

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Editorial note. Counted span is the author's own proof environment, source lines 1631-1640, following the statement at 1616-1629.

Editorial note. No omissions.

Allowed standard prerequisites (editorial).

- Closed immersion is etale-local on the target. Morphisms of Spaces, Lemma spaces-morphisms-lemma-closed-immersion-local (cross-chapter, cited at ~1635). It suffices to check X_i x_Y W -> W after a surjective etale W -> Y.
- Limits that are schemes. lemma-limit-is-scheme (same chapter, cited at ~1636). Reduces to the case of schemes.
- Scheme-level eventual closed immersion. Limits, Lemma limits-lemma-eventually-closed-immersion (cross-chapter, cited at ~1637-1638). Concluding engine for schemes.

Source reference. Original lemma 07SR: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 1435–1435. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0A0T: focused-seeds/geometry/S9-spaces-limits-eventually-closed-immersion/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 03M4: focused-seeds/geometry/S9-spaces-limits-eventually-closed-immersion/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let Y be an algebraic space over S. Let $X = \lim X_i$ be a directed limit of algebraic spaces over Y with affine transition morphisms. Assume*

(1) *Y quasi-compact and quasi-separated,*
(2) *X_i quasi-compact and quasi-separated,*
(3) *the transition morphisms $X_{i'} \to X_i$ are closed immersions,*
(4) *$X_i \to Y$ locally of finite type*

Date: stacks-project:spaces-limits.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

(5) $X \to Y$ *is a closed immersion.*

Then $X_i \to Y$ *is a closed immersion for* i *large enough.*

Proof. Choose an affine scheme W and a surjective étale morphism $W \to Y$. Then $X \times_Y W$ is a closed subspace of W and it suffices to check that $X_i \times_Y W$ is a closed subspace W for some i (Morphisms of Spaces, Lemma 03M4). By Lemma 07SR this reduces us to the case of schemes. In the case of schemes it follows from Limits, Lemma 0A0T. □